PHY F376: A study of hybrid classical-quantum algorithms

Week 7 (2/10/2026) - Validating CCSD-Seeded LUCJ: A Discrepancy Found Before Hardware

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Overview of This Week's Work

This week Prof. Indrakshi supplied CCSD t_1, t_2 amplitudes derived from the same H2O CAS(4e,4o) system, along with instructions to seed the LUCJ ansatz from these amplitudes instead of random parameters, validate the result on the noiseless simulator, and only then proceed to hardware. Most of this week went into implementing that validation step carefully, since it did not pass, and into narrowing down why.

Implementing the Validation

Following the supplied recipe exactly, I built the CCSD-seeded LUCJ circuit using `ffsim.UCJOpSpinBalanced.from_t_amplitudes` with the given amplitudes and interaction pairs, prepared the Hartree-Fock reference, and measured the noiseless variational energy. I also added a control case seeding with all-zero amplitudes, which should reduce exactly to the Hartree-Fock energy regardless of circuit details, as a check on the surrounding construction before trusting any result involving the real amplitudes.

What the Validation Revealed

The zero-amplitude control matched the Hartree-Fock reference exactly, confirming that the circuit construction and qubit-ordering handling are correct on their own. However, the CCSD-seeded circuit did not come close to the claimed CCSD energy: against an expected error of about 0.09 mHa, the actual error was around 60 mHa with the recipe as given, and did not meaningfully improve even when I used the full, mathematically exact, untruncated amplitude decomposition rather than a restricted one. I also checked sign-flipped and transposed versions of the amplitudes, none of which closed the gap. Since the reference determinant matches exactly but the correlation structure encoded in the amplitudes does not reproduce the correct energy under any of these variations, my working hypothesis is a mismatch in how spatial orbitals are indexed between the supplied Hamiltonian and the supplied amplitude file, rather than a bug in the circuit-building code itself.

Readying the Integration, Without Using It Yet

Per the instructions, hardware execution should only follow a successful validation, so I have not proceeded past the simulator. I did, however, wire the CCSD-seeded ansatz into the existing SQD pipeline exactly as before, changing only the state-preparation step and leaving the Hamiltonian parsing and SQD post-processing untouched, so that the comparison can be run immediately once the ordering issue is resolved.

Concluding Remarks

This week's result is a validation failure rather than a working comparison, but I believe it is a useful one to have caught before spending QPU time on it. The next step is to verify the orbital ordering convention used to generate the amplitudes, since that is the most likely explanation I have been able to narrow it down to.